

Entangled four-dimensional multicomponent topological states from photonic crystal defectsXiao Zhang,^{1,*} Youjian Chen,^{1,2} Yuzhu Wang,¹ Yuhao Liu,^{1,3} Jun Yu Lin,⁴ Nai Chao Hu,⁵ Bochen Guan,⁶ and Ching Hua Lee^{7,8,†}¹*School of Physics, Sun Yat-sen University, Guangzhou 510275, China*²*Department of Physics and Astronomy, Stony Brook University, Stony Brook, New York 11794-3800, USA*³*Department of Physics, University of Chicago, 5720 South Ellis Avenue, Chicago, Illinois 60637, USA*⁴*Department of Physics, The Chinese University of Hong Kong, Hong Kong, China*⁵*Department of Physics, University of Texas at Austin, Austin, Texas 78712, USA*⁶*Department of Electrical and Computer Engineering, University of Wisconsin, Madison, Wisconsin 53706, USA*⁷*Institute of High Performance Computing, Singapore 138632, Singapore*⁸*Department of Physics, National University of Singapore, Singapore 117542, Singapore*

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Recently, there has been a drive towards the realization of topological phases beyond conventional electronic materials, including phases defined in more than three dimensions. We propose a versatile and experimentally realistic approach of realizing a large variety of multicomponent topological phases in two-dimensional (2D) photonic crystals with quasiperiodically modulated defects. With a length scale introduced by a background resonator lattice, the defects are found to host various effective orbitals of *s*-, *p*-, and *d*-type symmetries, thus providing a monolithic platform for realizing multicomponent topological states without requiring separate internal degrees of freedom in the physical setup. Notably, by coupling the defect modulations diagonally, we report the realization of “entangled” 4D quantum Hall (QH) phases which cannot be factorized into two copies of 2D QH phases, each described by the first Chern number. The structure of this nonfactorizability can be quantified by a classical entanglement entropy inspired by quantum information theory. In another embodiment, we present 4D *p*-orbital nodal lines in a nonsymmorphic photonic lattice, hosting boundary states with an exotic manifold. Our simple and versatile approach holds the promise of topological optoelectronic and photonic applications such as one-way optical fibers.

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Introduction. Quantum Hall systems [1–3], topological insulators [4–8], and Weyl semimetals [9–11] rank among the most intensely studied topics in condensed matter physics [12–15]. Each energy band is ascribed a topological index that mandates the presence of interesting boundary states and quantized responses. Soon after their discovery in electronic materials, analogs in electrical [16–23], phononic (acoustic) [24–29], photonic [30–32], or their hybrids [33] have been proposed and even realized experimentally. Although topological phases with nontrivial Chern invariants in principle exist in any even number of dimensions, physical space is limited by three dimensions (3D). Topological phases characterized by the second Chern number C_2 [4D quantum Hall (QH) systems [34]], as well as 4D nodal band structures, cannot be directly realized within 3D physical space [35], and have for a long time been believed to exist only in theory.

Recently, such 4D QH phases were realized in 2D artificial systems with additional synthetic dimensions. Proposals include unidirectional propagating waveguides [36,37], cold atoms [38–40], optical fibers of Weyl materials with a helical structure [41], and an artificially generated parameter space of a four-level system [42]. In the former two, synthetic

dimensions arise by identifying the aperiodic Hamiltonian with a Hofstadter lattice containing twice the number of dimensions [43–47]. The resultant fractal band structure can be characterized by the C_2 in the combined space of the physical and synthetic dimensions. Experiments have been successful in realizing such quasiperiodic topological phases, with topological pumping by the first Chern number (C_1) observed in propagating waveguides [48], and pumping by the second Chern number (4D QH analog) recently reported in photonic [37] and cold atom systems [40].

So far, however, all theoretical and experimental quasiperiodic 4D QH systems in the literature have been restricted to 2D tensor products of two 1D Hofstadter models. As such, the 4D QH states obtained were always factorizable into a product of 2D QH states, whose properties are already well known. Specifically, the spectra of such 4D tensor product states are just the sum of two 2D QH spectra, whose spectral flow can always be understood via iterated first Chern number polarization [49–51].

As such, our goal is to propose experimentally realistic setups that host truly nontrivial 4D QH modes that cannot be understood via the first Chern number, i.e., *entangled* 4D quantum Hall modes with unremovable classical entanglement between the two 2D subsectors that they exist in. By computing the entanglement spectrum of various cuts, as defined in analogy to quantum entanglement, insights can be gleaned about

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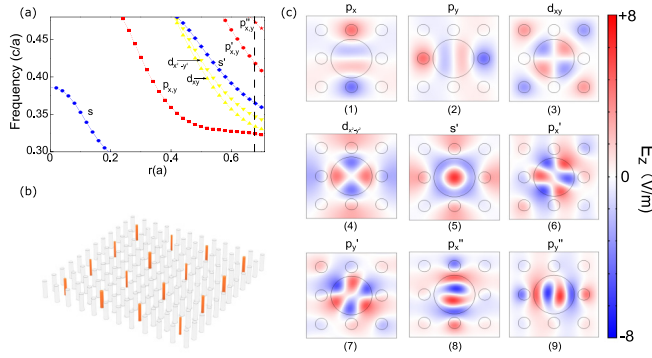


FIG. 1. (a) Dependence of the on-site frequencies of various s -, p -, and d -type defect modes with the defect rod radius r . Only s -type modes exist for $r < 0.2a$, the background rod radius, but progressively more exotic modes appear at larger radii. (b) Illustration of photonic crystal (PC) with background/defect rods colored gray/orange. Slight spatial modulation of the defect radii leads to an emergent entangled 4D QH state described in Fig. 2. (c) The E_z distributions of the various modes in (b). The proliferation of these modes leads to interesting multicomponent effective Hamiltonians.

how the subsectors are coupled. A fundamental observation is that the Wannier center response of entangled 4D systems cannot be expressed in terms of separate 2D Hall responses, as commonly taken for granted in the higher-dimension QH literature [38,52,53]. Our physical setup consists of a regular 2D background photonic crystal (PC) interspersed by defect resonators possessing aperiodically modulated radii. With their mature fabrication technology [54–63], such PCs can be easily fabricated to possess arbitrarily modulated defects, with practical applications such as one-way optical fibers. This freedom in modulation is not only crucial for realizing the (classically) entangled 4D QH modes, but also yields higher p, d effective orbitals that leads to multicomponent phases far beyond the conventional Hofstadter model, such as 4D p -orbital nodal lines in a nonsymmorphic photonic lattice.

Tight-binding Hamiltonian of a defect lattice. Our photonic crystal (PC) contains a background lattice of equally spaced identical dielectric rods separated by air. Importantly, embedded in the PC are “defect” rods of quasiperiodically modulated radii. As shown in Fig. 1, these defect rods act as resonators that host confined transverse magnetic (TM) electromagnetic modes, which serve as photonic analogs to the atomic orbitals of a conventional material. To identify our PC with a quasiperiodic topological lattice, we express Maxwell’s equation in the basis of the TM defect modes, such that it plays the role of the Hamiltonian in the time-dependent Schrödinger’s equation [64,65]

$$H|\psi\rangle = \begin{bmatrix} 0 & i\varepsilon_0^{-1}\vec{\nabla}\times \\ -i\mu_0^{-1}\vec{\nabla}\times & 0 \end{bmatrix} \begin{bmatrix} \vec{E} \\ \vec{H} \end{bmatrix} = \omega|\psi\rangle, \quad (1)$$

where H is the effective TB Hamiltonian, $|\psi\rangle = [\vec{E}]$ denotes an eigenmode with eigenfrequency ω , and ε_0 is the spatially varying dielectric constant. We shall always set the background rods to have relative dielectric constant 8 and radius

$0.2a$, where the lattice spacing a is defined as the distance between adjacent defect rods.

The shape of a defect mode depends qualitatively on the defect rod radius. For different radii, the defect modes can exhibit s -, p -, or d -type symmetries [66] (Fig. 1 and Ref. [67]), hence providing a monolithic platform for constructing multicomponent topological states without the need for multiple physical “orbitals.” As illustrated in Fig. 1, even with defect rods (central circles) of circular cross section, various defect modes with s , p , or d symmetries can be realized at sufficiently large rod radii. For these modes to robustly serve as topological degrees of freedom, we require them to lie within the frequency gap of the background resonators, which from our simulations [67] range from $\omega \approx 0.32c/a$ to $\omega \approx 0.46c/a$, with c being the speed of light. Within this frequency window, defect rods thinner than the background rods ($r < 0.2a$) can support only s -type orbitals [Fig. 1(a)]. However, thicker defect rods can support not just p_x - or p_y -type orbitals, but also simultaneously more exotic $d_{x^2-y^2}$, d_{xy} , p'_x , p'_y , etc., orbitals [see Supplemental Material (SM) [67] for their field distributions] at a sufficiently large radius. It is only through the length scale introduced by the background rods that we obtained very localized qualitatively distinct s , p , and d modes.

For simplicity, we shall mostly focus on multicomponent Hamiltonians spanned by only s - and p -type defect modes, which respectively arise from defect rods with radii $r < 0.2a$ and $r > 0.2a$. Due to their locality, uncoupled s - and p -type defect modes form nearly flat bands with eigenfrequencies almost perfectly proportional to the defect radius (see SM [67]). As such, they collectively give rise to an effective tight-binding (TB) Hamiltonian [64,68] with on-site energies ω_i proportional to the radius of the i th rod, and almost constant hopping amplitudes t_{ij} between nearest-neighbor orbital sites $|\psi_i^\alpha\rangle, |\psi_j^\beta\rangle$,

$$H = \sum_i \sum_{\alpha \in s, p, d} \omega_i |\psi_i^\alpha\rangle \langle \psi_i^\alpha| + \sum_{(i, j)} \sum_{(\alpha, \beta) \in s, p, d} t_{ij}^{\alpha\beta} |\psi_i^\alpha\rangle \langle \psi_j^\beta|. \quad (2)$$

Further hoppings can be neglected as the defect modes are localized. For s -type modes, it suffices to approximate t_{ij}^s by a constant t , whereas for p modes, $t_{ij}^p \approx t_\sigma$ or t_π depending on whether σ or π bonding is involved.

Entangled 4D QH modes. Since the topological pumping of 2D QH states can be realized in the 1D lines of the defect discussed above, we should be able to probe 4D QH states in 2D defect lattices [Fig. 1(b)] with two synthetic dimensions. A 4D QH state is characterized by a nonzero

$$C_2 = \frac{1}{8\pi^2} \int F \wedge F d^4k \\ = \frac{1}{4\pi^2} \int F_{zx}F_{yw} + F_{xy}F_{zw} + F_{yz}F_{xw} d^4k, \quad (3)$$

where $F_{\mu\nu} = \langle \partial_{k_\mu} \phi | \partial_{k_\nu} \phi \rangle - \langle \partial_{k_\nu} \phi | \partial_{k_\mu} \phi \rangle$ is the (2D) Berry curvature for a state ϕ which also gives rise to a first Chern number $C_1^{\mu\nu} = \frac{1}{2\pi} \int F_{\mu\nu} d^2k$. Due to its quadratic dependence on $F_{\mu\nu}$, the 4D QH effect possesses interesting nonlinear response properties [38] and gives the parent state for various descendant 2D and 3D topological states [69].

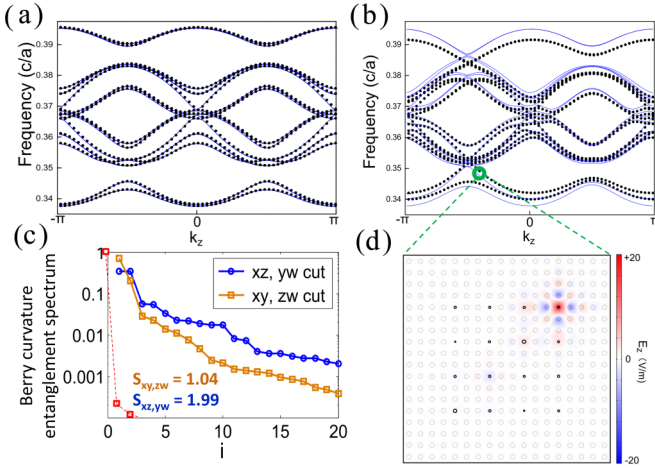


FIG. 2. (a) PBC and (b) OBC frequency dispersion of the entangled defect lattice of Eq. (4), with parameters $r_0 = 0.1a$, $r_1 = r_2 = 0.025a$, $k_w = 0$, and rational flux $b = 1/4$. Excellent numerical agreement exists between the FEM simulation/TB model [Eq. (5) with $n = 4$ along x and y ; by fitting the band data with Eq. (5), we found that $\omega_0 = 0.368\,62(c/a)$, $\lambda = -0.014\,508$, and $t = -0.003\,98(c/a)$] results, which are represented by the dotted/continuous curves. With OBC, topological spectral flow exists between the central and top/bottom defect bulk bands. (c) The Berry curvature entanglement spectra $\tilde{\lambda}_i$ and entropy for both xy, zw and xz, yw cuts, with long tails indicative of nontrivial entanglement, i.e., nonfactorizability of the second Chern number. An unentangled defect lattice given by r_{ind} gives a delta-function-like spectrum (red). (d) The E_z distribution of the midgap mode circled green [$k_z = -1.2$, $\omega = 0.3465(c/a)$] in (b), which is clearly localized at a corner defect rod (thick circle).

The simplest approach to a photonic analog of a 4D QH state is to consider independent defect modulations in both directions, i.e., $r_{\text{ind}}(x, y) = r_0 + r_1 \cos(2\pi bx + k_z) + r_2 \cos(2\pi by + k_w)$ (see SM [67] for detailed results), as is performed with cold atoms in the literature [45,48]. However, a key limitation of this approach is that its resultant eigenmodes are just *product states* of two first Chern number eigenmodes. Therefore, the topological pumping of such a product 4D QH mode can be completely understood in terms of the first Chern number, despite formally possessing a second Chern number invariant.

To obtain a truly nontrivial 4D QH state analog, the key insight is to instead consider *entangled* modulations, the simplest of which takes the form

$$r(x, y) = r_0 + r_1 \cos[2\pi b_z(x + y) + k_z] + r_2 \cos[2\pi b_w(x - y) + k_w], \quad (4)$$

where r_0, r_1, r_2 are constants, and x, y labels the defect sites in the $\hat{x}, \hat{y}$ direction [Fig. 2(d)]. Analogous to the 1D Aubry-André-Harper (AAH) model [67], b_z and b_w are flux parameters conjugate to synthetic momenta k_z and k_w . Notably, this system is not factorizable into a direct product of two AAH models, despite the seemingly naive replacement $(x, y) \rightarrow (x + y, x - y)$. To concretely see why, we examine its effective

TB Hamiltonian

$$H_{4D} = \sum_{x,y} \omega_0 + \lambda \cos[2\pi b_z(x + y) + k_z] w |\psi_{(x,y)}\rangle \langle \psi_{(x,y)}| + \lambda \sum_{x,y} \cos[2\pi b_w(x - y) + k_w] |\psi_{(x,y)}\rangle \langle \psi_{(x,y)}| + t \sum_{x,y} (|\psi_{(x,y)}\rangle \langle \psi_{(x+1,y)}| + |\psi_{(x,y)}\rangle \langle \psi_{(x,y+1)}| + \text{H.c.}), \quad (5)$$

possessing t hoppings in the $\hat{x}$ and $\hat{y}$ directions which are *not* aligned with the phase factors of $2\pi b(x \pm y)$ along the k_z and k_w directions. In other words, while the effective Landau gauge is taken with respect to the $\frac{\pi}{4}$ -rotated directions $\hat{x} \pm \hat{y}$, the physical positions of the defect rods are still arranged in a lattice spanned by $\hat{x}$ and $\hat{y}$ basis vectors. As such, Eqs. (4) and (5) represent a 4D QH Hamiltonian with nontrivial entanglement between its two AAH subsystems. The frequency spectrum of the defect modes can be computed by a finite-element Maxwell's equation (FEM) solver, and is presented, for $b = 1/4$, in Figs. 2(a) and 2(b) for s -type defect modes with a periodic/open boundary condition (PBC/OBC) in the $\hat{x}, \hat{y}$ direction. Upon varying k_y over a period, boundary modes in the OBC case [Fig. 2(d)] are seen to continuously connect the bulk bands. This spectral flow of midgap states is a signature of topological behavior, and can indeed be understood via the effective TB Hamiltonian for s -type orbitals.

This (classical) entanglement within H_{4D} can be quantified in analogy to quantum entanglement. Generically, we define an entanglement cut by decomposing a given state space Σ as a product of two desired subspaces: $\Sigma = \Sigma_1 \otimes \Sigma_2$. For a particular state ϕ , how much the degrees of freedom (DOFs) of the two subsystems Σ_1 and Σ_2 are entangled can be expressed through the singular-value-decomposition (SVD)

$$\phi = \sum_i \lambda_i \phi_1^i \otimes \phi_2^i, \quad (6)$$

where ϕ_α^i , $\alpha = 1, 2$ is nonzero only within Σ_α . $\lambda_1 \geq \lambda_2 \geq \lambda_3 \dots$ are known as the entanglement eigenvalues, and are normalized according to $\sum_i \lambda_i = 1$. In the special case of ϕ being a product state, there will only be one nonzero λ_i , i.e., $\lambda_1 = 1$ and $\lambda_{i \geq 2} = 0$. The amount of entanglement, or extent of departure from a product state, is measured through the *entanglement entropy*

$$S = - \sum_i \lambda_i \log \lambda_i \geq 0, \quad (7)$$

through which the rank, or effective number of independent DOFs, is given by $e^S \geq 1$.

Whether the second Chern number C_2 is trivially the product of two C_1 's (as in previous literature) depends crucially on the factorizability of the integrand in Eq. (3). In the purely factorizable case dictated by r_{ind} , the independence of the x, k_z DOFs from the y, k_w DOFs implies that only the $F_{zx}F_{yw}$ term is nonzero, and that $C_2 = \frac{1}{(2\pi)^2} \int F_{zx} dk_x dk_z \int F_{yw} dk_y dk_w = C_1^{zx} C_1^{yw}$. But with the two AAH subsystems entangled as in Eq. (4), the integrand for C_2 becomes nonfactorizable, albeit SVD decomposable into a linear combination of factorizable terms. Hence we can define the *Berry curvature entanglement*

spectrum $\{\tilde{\lambda}_1, \tilde{\lambda}_2, \dots\}$ and (classical) *entanglement entropy* $S_{xz,yw}$ of a 2D defect lattice (H_{4D}) via Eqs. (6) and (7), with ϕ being the integrand of C_2 and the cut demarcating the two coupled AAH subsystems with x, k_z and y, k_w DOFs.

As plotted in Fig. 2(c), H_{4D} possess a long tail of Berry curvature entanglement eigenvalues representing not one but $e^{S_{xz,yw}} \approx 7.3$ effective entangled DOFs. Also presented in Fig. 2(c) is the entanglement $S_{xy,zw}$ between the spatial (x, y) and magnetic translation (k_z, k_w) DOFs, which possess a double degeneracy in the largest few eigenvalues. Interestingly, the Berry curvature entanglement eigenvalues $\tilde{\lambda}_i$ can be expressed in terms of the entanglement eigenvalues [70] λ_i of the *defect band* as well as its SVD-decomposed Berry connection and curvature, as detailed in the SM [67].

Since the defect lattices are quasiperiodic, the Berry curvatures are most directly computed by implementing the momentum conjugate to the quasiperiodicity as a threaded flux through an edge [71] (see SM [67]). With the defect lattice given by Eq. (4), the lowest two defect bands intersect (Fig. 2), giving rise to a *non-Abelian* second Chern number $C_2 = \frac{1}{4\pi^2} \int F_{zx}(k)F_{yw}(k) + F_{yz}(k)F_{xw}(k) + F_{xy}(k)F_{zw}(k) d^4k = 1 + 0 + 1 = 2$.

This expression for C_2 , which interprets the 4D electromagnetic response as iterated Hall responses with the Berry flux and magnetic field [67], suggests an important physical consequence of entanglement to Wannier center evolution, the electromagnetic pumping of a maximally localized occupied state. As elaborated in the SM [67], the Wannier response coefficient in the generic *entangled* case is $\frac{1}{(4\pi^2)^3} \int F_{zx}(k)F_{yw}(k') + F_{xy}(k)F_{zw}(k') + F_{yz}(k)F_{xw}(k') d^4k d^4k'$, a much more complicated expression that does not reduce to C_2 . The discrepancy arises from couplings between the two QH copies which introduce 4D dispersion to the Berry flux.

4D nodal lines with nonsymmorphic symmetry. The propensity for our photonic setup in realizing entangled phases extends beyond QH states. Consider the rather extreme but perfectly realistic case with rod “defects” being their anisotropy. As detailed below, various interesting 4D nodal line phases can arise when the modulations and orientations of these elliptical rods obey nonsymmorphic (glide reflection) symmetry, which has very much been associated with exotic topological phenomena [72] in materials and metamaterials [32,73–75].

For concreteness, consider the photonic unit cell of Fig. 3(a). It consists of eight 12° tilted rods A–H with equal eccentricities, grouped into four subcells with position labels $(x, y) = (0, 0), (0, 1), (1, 0), (1, 1)$. An effective 4D system emerges from synthetic DOFs k_z, k_w which tune the rods’ major axes according to their subcell positions, as manifested from on-site energies $H_{\text{on-site}} \propto \cos(\pi x + k_z) + \cos(\pi y + k_w)$ (see SM [67] for the tight-binding model). In particular, the nonsymmorphic pg symmetry $g_y = \{m_y | \tau_x\}$ exists in the 3D planes $k_z = \pm\pi/2$, since rods A–D and rods E–H will have equal major axes, respectively, and hence respect the nonsymmorphic condition $g_y \psi(x, y) = \psi(x + \frac{a}{2}, -y)$.

In this quasi-4D system, line degeneracies, i.e., nodal lines of the p_x, p_y bands, emerge, for instance, in the k_x – k_w plane at $k_y = 0, k_z = \pi/2$ [inset of Fig. 3(b)]. In certain synthetic momentum slices, i.e., $k_w = \pi/2$, the degeneracy becomes a

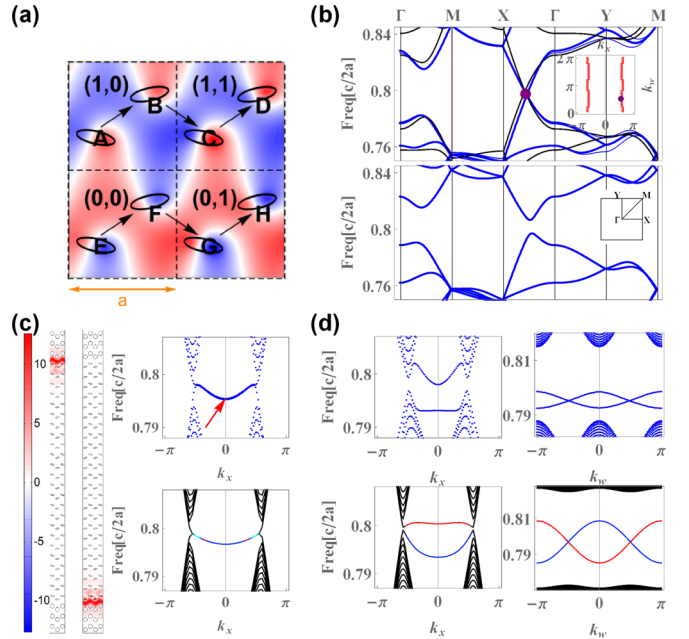


FIG. 3. (a) Unit cell of our effectively 4D PC with two synthetic dimensions, with red/blue representing positive/negative electric fields. Nonsymmorphic symmetry holds when $k_z = \pm\pi/2$, as indicated by arrows that map the elliptical rods to themselves via the glide reflection g_y . (b) Bulk band structure from COMSOL simulations (blue) and TB model (black) in the $k_z = k_w = \pi/2$ slice, with nonsymmorphic symmetry preserved/broken by tilting half of the rods (top/bottom). Top: A Dirac point degeneracy (purple dot) lies on nodal lines (red) in the 4D BZ, as plotted in the k_x – k_w slice with $k_z = \pi/2, k_y = 0$ (inset). Bottom: Degeneracies are all gapped by breaking nonsymmorphic symmetry. (c) Degenerate boundary states in the $k_z = k_w = \pi/2$ slice under y -direction OBCs. Left: Spatial profile of boundary degeneracies at $k_x = 0$, marked by the red arrow in the COMSOL band structure (top right), which corroborates with TB results (bottom right). (d) Boundary modes that do not merge into the bulk, shown for slices $k_z = k_w = \pi/2$ (left) and $k_x = 0, k_z = \pi/2$ (right). Simulations (top) qualitatively agree with TB results (bottom), with black and red/blue labeling bulk and left/right edge modes, respectively.

Dirac point, as illustrated by the purple dot in both Fig. 3(b) (upper panel) and its inset. The crucial role of nonsymmorphic symmetry g_y can be seen by tilting rods B, D, F, H, i.e., to 30° , which gaps out the line degeneracies [Fig. 3(b), lower panel]. Such phase transitions crucially rely on elliptical rods that are easily engineered in our photonic setup, but are impractical in material realizations.

Another interesting aspect of our 4D nodal system, as compared to its 3D counterparts, is that topological boundary states can be pumped without merging into the bulk. Unlike “drumhead” surface states of 3D systems which do not extend throughout the whole Brillouin zone (BZ), our surface states are only bounded by “slices” of the nodal boundary projection, and can remain separated from the bulk across the whole synthetic parameter period [36,37]. This is shown in Fig. 3(d) with open boundaries in y (see SM [67] for more slices).

Experimental prospects. With current advances in fabrication technology [76–78], our proposed entangled modes can

be experimentally observed relatively simply. We propose a 2D PC [Fig. 1(b)] consisting of a background lattice of rods of radius $0.2a$ and dielectric constant 8, with a in the decimeter range. Every third rod is replaced by an otherwise identical defect rod of radius determined by Eq. (4). The spectral flow of the edge modes can be measured when synthetic dimension parameters k_z, k_w are varied over a period. This edge mode exists in the bulk gap, and will appear as a distinct peak in the microwave transmission spectrum [56,58,59] when the wavelength is tuned within the bulk gap. With our entangled 4D QH mode, a resonant peak is, for instance, expected at the wavelength $0.2833m$, i.e., when $k_z = -1.37$ and $k_w = 0$ (Fig. 1). Alternatively, the edge mode can also be probed via photonic topological boundary pumping with paraxial (out-of-plane) light [37].

Conclusions. We have proposed a very versatile and realistic way of realizing analogs of generic multicomponent gapped QH and gapless nodal topological states through spatially modulated 2D photonic lattices. Modes around the defect rods are governed by Hofstadter Hamiltonians, and

consequently behave as QH wave functions. With quasiperiodic modulations coupling the two directions, we obtain entangled 4D QH modes and nodal lines/points protected by nonsymmorphic symmetry. Such is the versatility of our PC platform that by simply modulating the rod size/shapes, existing PC experiments on “designer” disorder, non-Hermitian [79], high-order [80], interacting [81], and optomechanical [33] topological/nontopological effects can be naturally generalized to higher dimensions. Applications include one-way optical fibers, with topologically protected chirality in the space of in-plane/out-of-plane modes.

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